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# Evaluating the role of non-HIV antigens in the initiation of QA013.2 HIV bnAb lineage

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**Abstract**

Broadly neutralizing antibodies (bnAbs) are a major goal of HIV vaccine design because they neutralize diverse viral strains. Breadth is typically achieved via affinity maturation and accumulated somatic mutations. However, the inferred naïve precursors of some bnAbs fail to bind the HIV epitopes targeted by their mature counterparts. Consequently, the initiating antigens remain elusive, complicating vaccine design. Building on our previous identification of the HIV bnAb QA013.2 and evidence that its lineage acquired somatic mutations prior to HIV seroconversion, we explored the hypothesis that QA013.2 maturation may be initiated by non-HIV antigens. We screened the inferred naïve precursor and a pre-HIV mutated intermediate of this lineage against > 114.000 antigens, including microbial and parasite glycans, the human proteome and peptides from the human virome. Although human proteome profiling yielded reproducible signals for seven candidates, these interactions lacked a defining structural motif or clear context for B cell priming. Across all platforms, no definitive initiating antigen was identified under the conditions tested, suggesting that if early maturation is indeed driven by non-HIV exposure, the required antigen may involve low-affinity interactions or complex conformational features or glycosylation patterns that remain challenging to detect in standard high-throughput screens.

## Introduction

Developing an effective HIV vaccine remains a global priority, with an estimated 1.2 million new infections occurring annually[1]. These efforts are challenged by the extraordinary diversity of HIV and its rapid mutation rate. Broadly neutralizing antibodies (bnAbs) are a major focus of vaccine design, as they can recognize diverse viral strains and passive immunization trials have demonstrated their capacity to confer protection in non-human primates[2–7]. BnAbs typically arise only after years of infection, and notably do so following superinfection and require extensive somatic hypermutation[8–10]. To date, no vaccine has successfully induced mature HIV bnAbs.

An intriguing feature of several potent bnAb lineages including VRC01, 2G12, 2F5, PG9, 447-52D and PGT121 is that there is no detectable binding of the inferred naïve precursors to the cognate HIV antigen recognized by their mature counterpart[11–17]. This presents a major challenge for vaccine design, as it makes it difficult to develop immunogens capable of engaging precursor B cells. Thus, there is a need to identify the antigens that

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engage the naïve and early members of HIV bnAb lineages, as these interactions may provide a template for improved vaccine design.

We previously identified QA013.2[18], an HIV bnAb whose inferred naïve precursor does not bind the autologous initial infecting viral envelope recognized by the mature antibody[19]. QA013.2 was isolated from an individual following HIV superinfection[20], 2,282 days after initial infection[18]. The mature antibody displays hallmark features of V3-directed HIV bnAbs, including dependence on the N301 and N332 glycans and recognition of the linear GDIR motif. Longitudinal PBMC samples spanning pre- and post-HIV infection enabled reconstruction of the QA013.2 lineage and inference of its naïve precursor[18,19]. Using bulk sequencing, we previously identified clonally related variable heavy (VH) and light (VL) chain sequences across this lineage and found, unexpectedly, that most VH sequences were sampled before HIV infection. Intriguingly, early members of this lineage were detected prior to HIV seroconversion. These pre-infection sequences already showed somatic hypermutation (SHM) relative to the inferred naïve precursor antibody and shared mutations with the mature bnAb, suggesting that early maturation of the QA013.2 lineage may be initiated before HIV infection (Fig. 1). Consistent with this, the inferred naïve precursor antibody did not show detectable binding to envelope protein of the initial infecting strain[19]. These data raise the possibility that non-HIV antigenic stimulation may have initiated or shaped QA013.2 early maturation.

We hypothesized that non-HIV antigens, such as glycans, human proteins, or other pathogens, may have primed the naïve QA013.2 precursor, which later evolved into a cross-reactive antibody and ultimately into a bnAb (Fig. 2). Supporting this idea, some HIV-specific antibodies cross-react with non-HIV antigens, including human proteins, commensal bacteria and pathogen-derived glycans, and some antibodies capable of recognizing HIV-1 have been identified in HIV-naïve individuals[21–27]. Together, these findings suggest that eliciting optimal HIV antibodies may require immunogens beyond HIV-derived antigens alone. This hypothesis has been difficult to test because samples from prior to HIV infection are rarely available and naïve sequence inference has limitations, especially in the absence of data from longitudinal sampling. The QA013.2 lineage offers a unique opportunity to address this question, as our unusually rich sequence dataset, including pre-HIV timepoints, enabled confident inference of the naïve precursor and early lineage members, in addition to identifying mutated sequences present prior to infection.

Here, we sought to identify candidate antigens capable of engaging the inferred naïve precursor of QA013.2 antibody. To this end, we used complementary high-throughput peptide, protein, and glycan screening platforms,

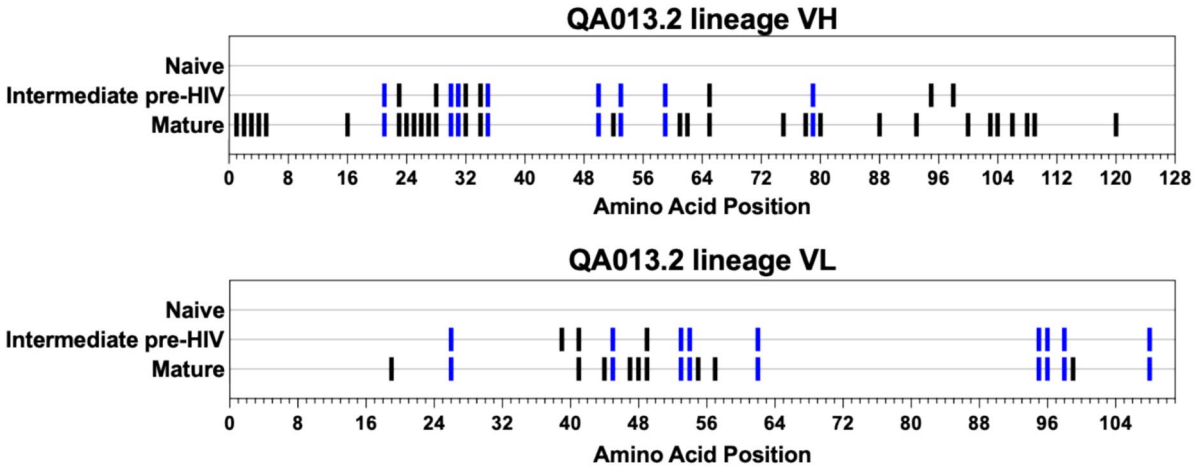



**Fig. 1** QA013.2 antibody lineage variable region amino acid substitutions relative to inferred naïve precursor. Heavy-chain (top) and light chain (bottom) amino acid sequence alignment of QA013.2 inferred naïve precursor, mature, and intermediate antibody sampled at a pre-HIV timepoint. Vertical lines indicate amino acid substitutions relative to the inferred naïve precursor. Black lines indicate substitutions that are unique to the indicated antibody, blue lines highlight substitutions shared between intermediate and mature antibodies.

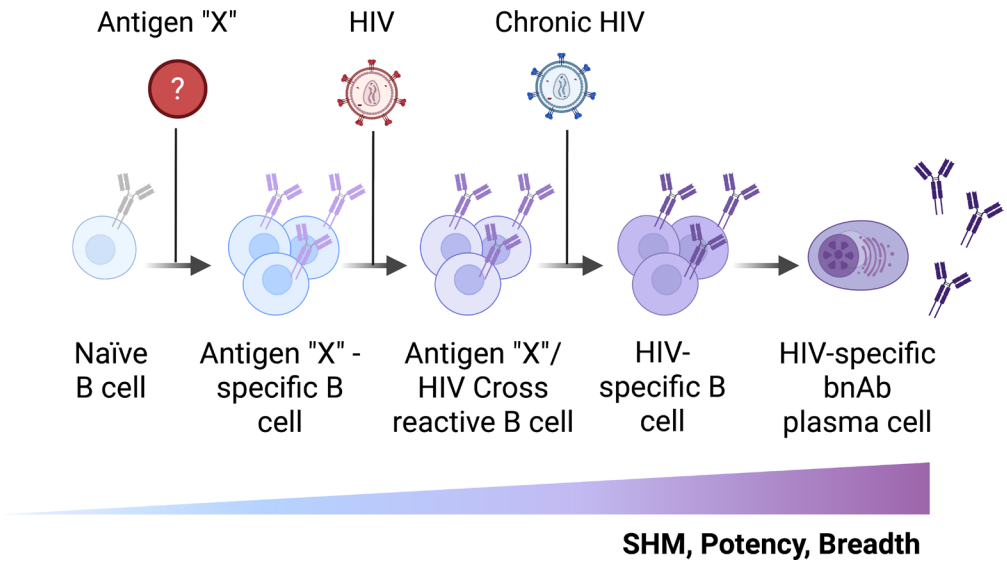



**Fig. 2** Hypothetical model for QA013.2 HIV bnAb development. A naïve B cell with a receptor specific to Antigen "X" is initially activated upon encountering the antigen. Upon activation, the naïve B cell becomes antigen specific and proliferates, generating a pool of antigen "X"-specific B cells. With exposure to HIV, B cells that are cross-reactive between antigen "X" and HIV are selected for. During chronic HIV infection, these cross-reactive B cells undergo somatic hypermutation (SHM) and affinity maturation, becoming increasingly specific to HIV antigens. Following further rounds of SHM and selection, the B cells develop into plasma cells capable of producing broadly neutralizing antibodies.

including VirScan, HuProt, and microbial and parasite glycan arrays, together representing more than 114,300 antigens.

## Results

### Microbial glycan microarray did not detect glycan reactivity of QA013.2 lineage

Because the mature QA013.2 antibody recognizes the HIV-1 V3-glycan supersite and prior work has shown that HIV inferred naïve precursor can bind to glycopeptides[28], we tested whether the QA013.2 inferred naïve precursor and pre-HIV intermediate antibodies exhibited glycan reactivity. Glycan binding was assessed using a Microbial Glycan Microarray, which is comprised of polysaccharides derived from 313 distinct Gram-negative and Gram-positive bacterial species[29]. As a positive technical control, serum samples were included and displayed robust and diverse glycan binding across the array, confirming the integrity and sensitivity of the platform (Fig. 3A). In contrast, the inferred naïve precursor QA013.2 antibody exhibited signal intensities that remained near baseline across all glycans, with no discrete peaks exceeding background levels (Fig. 3C). Similarly, an early intermediate of the QA013.2 lineage circulating 379 days prior to the estimated time of HIV infection, which differed by 27 amino acids from the inferred naïve precursor and showed limited binding to HIV and no detectable neutralization, lacked detectable glycan reactivity (Fig. 3D) as did an isotype control (Fig. 3B). No binding was observed for the inferred naïve precursor or early intermediate under the conditions tested, yielding no evidence of broad bacterial glycan recognition during early QA013.2 lineage maturation.

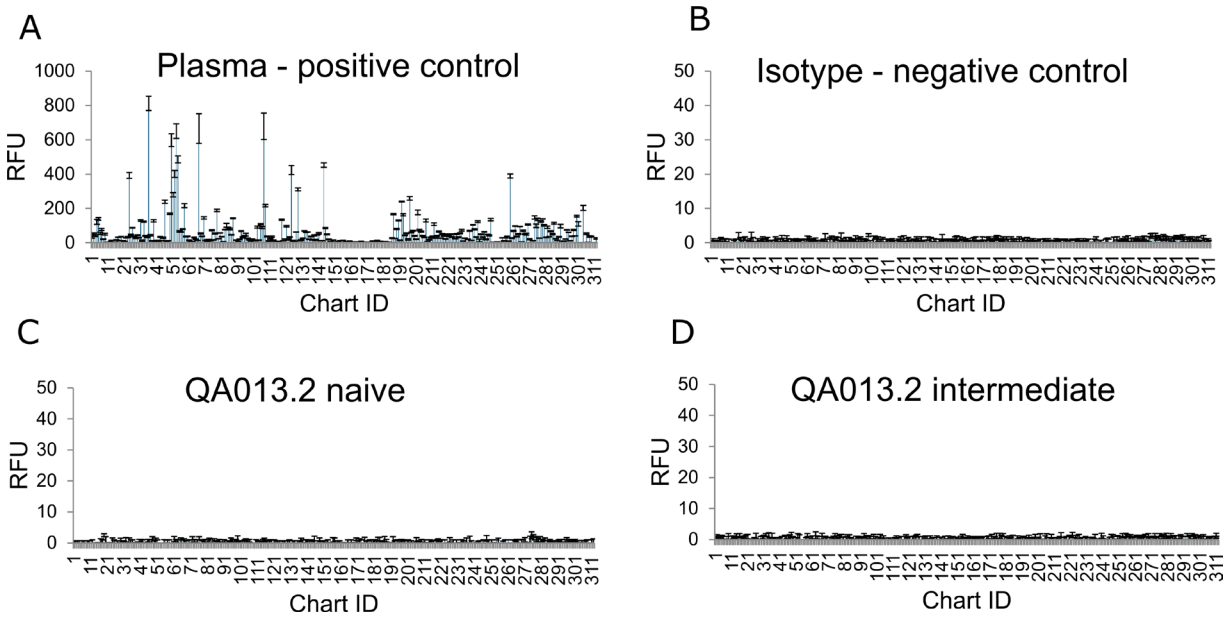



**Fig. 3** Binding assessment of QA013.2 antibody lineage to a microbial glycan microarray. Relative Fluorescence Units (RFU) indicate the binding intensity of antibodies to a diverse panel of microbial glycans and chart IDs refer to the glycan identity (TableS2). (**A**) Positive control plasma illustrating the magnitude of positive binding. (**B**) Isotype negative control (SARS-CoV-2 antibody C68.59 inferred naïve precursor) showing background binding. (**C**-**D**) Binding profiles of QA013.2 inferred naïve precursor and intermediate antibodies. Antibodies were used at 50 µg/mL. The highest and lowest values from each set of six replicates were removed to eliminate the false hits that contain a single very high or low point. Data are presented as mean RFU across technical replicates (n = 4), with error bars representing standard deviation.

### QA013.2 lineage antibodies do not bind schistosome antigens despite donor seropositivity

Because prior work has suggested that *Schistosoma mansoni* parasite glycans can engage some glycan-reactive HIV-1 bnAb lineages, such as 2G12, PGT121, PGT151, and PGT128 as well as PCDN76 and its unmutated ancestor[24], we next asked whether exposure to schistosomes might have contributed to QA013.2 lineage maturation. ELISA showed that plasma samples from the QA013.2 donor before and after HIV seroconversion were reactive to schistosome antigens, with post-HIV infection plasma timepoints exceeding pre-HIV infection levels (Fig. 4A). These data suggest that QA013 donor was exposed to schistosomes prior to HIV infection and mounted a strong antibody response to this parasite.

To test whether QA013.2 lineage could have been primed by schistosomes prior to HIV infection, we tested antibodies representing different stages of the QA013.2 lineage, including the inferred naïve precursor, an intermediate lineage member, and the mature antibody. Data showed that all QA013.2 antibody binding values were near background and below the positivity cutoff (Fig. 4A). Similar low reactivity was observed for the isotype control. As expected, positive control PGT121, which was previously shown to react with *Schistosoma mansoni* glycans, displayed detectable binding while negative control VRC01 showed no binding[24].

Because the ELISA kit only included limited *Schistosoma* soluble egg antigens, we extended our study to a schistosome-based glycan array made of 32 compounds (glycans and glycopeptides, and undefined fractions from *Schistosoma* eggs). As a positive technical control, lectin cocktail samples were included and displayed robust binding across the array, confirming the integrity and sensitivity of the platform (Fig. 4B). In contrast, the inferred naïve precursor of the QA013.2 antibody exhibited signal intensities that remained near baseline across all glycans (Fig. 4D), with no discrete peaks exceeding background levels, on par with the isotype negative control (Fig. 4C). Similarly, an early intermediate of the QA013.2 lineage and the mature QA013.2 bnAb also lacked

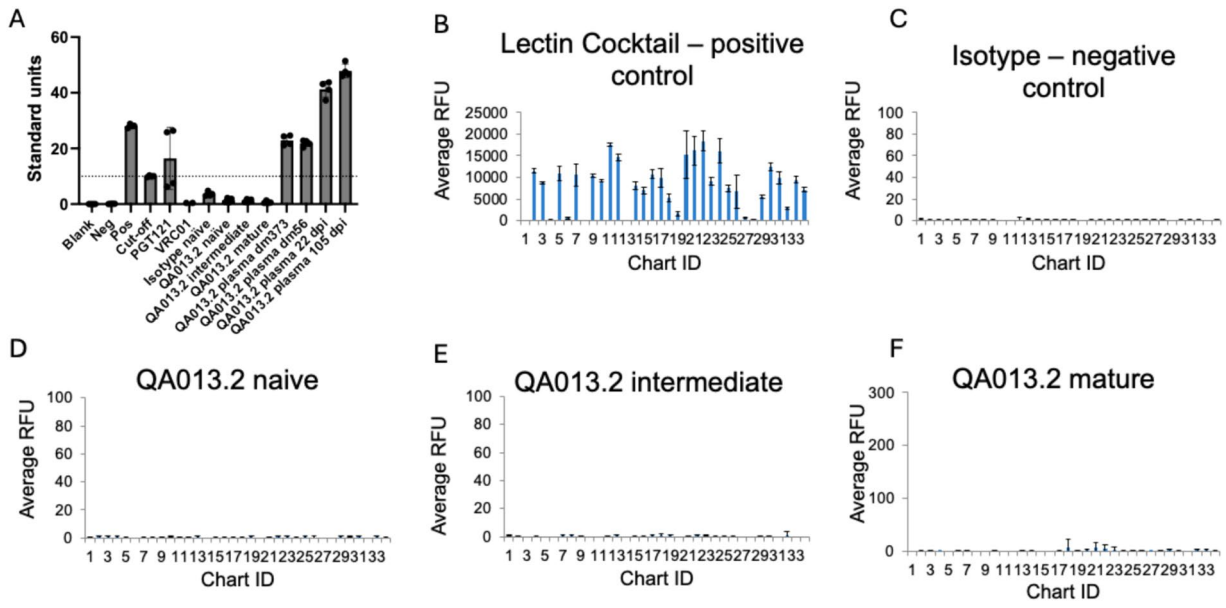



**Fig. 4** Assessment of QA013.2 antibody lineage binding to schistosome proteins and glycans. (**A**) ELISA detecting human IgG binding to *Schistosoma mansoni* egg antigens. The dashed line indicates the kit-defined positivity cutoff (10 standard units). Samples include negative and positive controls provided in the ELISA kit (ab108769), reference antibodies: PGT121 (HIV V3-glycan specific; positive control, previously shown to cross-react with Schistosoma) and VRC01 (HIV CD4-binding site specific; negative control, non-reactive with Schistosoma), QA013.2 lineage antibodies (inferred naïve precursor, intermediate, mature), and longitudinal QA013.2 plasma samples (days post-infection (dpi), days minus (dm)). Plasma samples were diluted to 1:100, and antibodies were used at 50 µg/mL. Each sample was tested in technical duplicate across two independent experiments (n = 4 total measurements). Data are presented as the mean ± SD of the standard-unit values from the four measurements. (**B**-**F**) Schistosome glycan microarray analysis. Bar plots show average fluorescence intensity (RFU) for (**B**) lectin cocktail positive control (ConA 10 µg/mL, AAL 1 µg/mL, and WFA 10 µg/mL, which recognize high-mannose, fucosylated, and terminal N-acetylgalactosamine (GalNAc) glycans, respectively), (**C**) isotype negative control (SARS-CoV-2 antibody C68.59 inferred naïve precursor), and QA013.2 lineage antibodies: (**D**) inferred naïve precursor, (**E**) intermediate, and (**F**) mature. Antibodies were used at 50 µg/mL. Each glycan is printed in quadruplicate (4 spots per glycan), and values represent the mean ± SD of these replicate spots. X-axes represent glycan ID (ChartID in TableS3).

detectable reactivity (Fig. 4E-F). These data indicate that although the QA013.2 donor was seropositive for schistosomes, the QA013.2 lineage itself did not exhibit detectable binding to schistosome antigens in this assay.

## VirScan profiling did not detect heterologous viral peptides engaging the QA013.2 lineage

To test the hypothesis that antigens derived from other pathogens might prime the QA013.2 lineage, we used the phage display library VirScan, which contains ~93,000 linear peptides from viruses with human tropism. As a benchmark, we first examined the VirScan profile of the HIV-1 V3-directed antibody 447-52D[30]. Using an unadjusted significance cutoff, we identified 475 peptides enriched in 447-52D compared to the library input (nominal p values; $p < 0.05$ in both replicates), of which 38 were derived from HIV-1 (Fig. 5A, TableS4). Following Benjamini–Hochberg correction for multiple hypothesis testing, 19 peptides remained significantly enriched, all of which were HIV-specific (TableS7). By contrast, while we identified 409 hits for negative isotype control prior to multiple testing correction (Fig. 5E), none remained significant after adjustment (TableS4).

We next assessed the reactivity of inferred naïve precursor, intermediate, and mature QA013.2 antibodies. Prior to multiple hypothesis testing correction, these antibodies enriched 353, 437, and 375 peptides, respectively ($p < 0.05$ in both replicates) (Fig. 5B-D, TableS4), though none remained significant following adjustment. Nonetheless, to rule out possible low-level binding that we were missing with the more stringent adjusted analysis, we examined overlap across these three conditions, which revealed 12 shared peptides (TableS6). No conserved

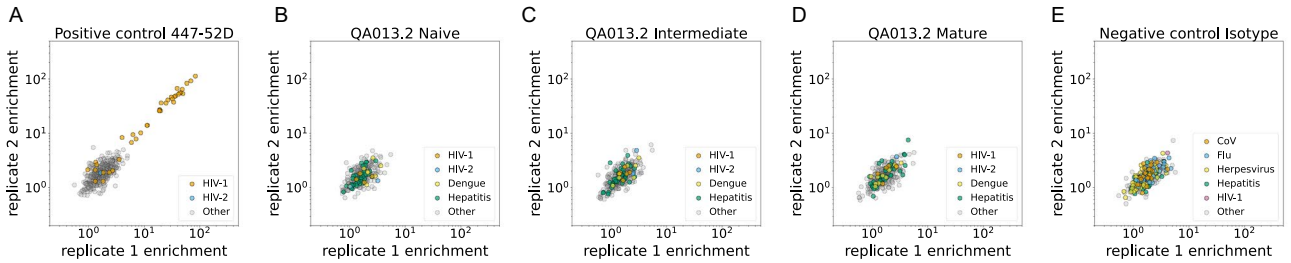



**Fig. 5** Profiling of QA013.2 antibody lineage reactivity against the human virome. Enrichment of human viral linear peptide targets was assessed by PhIP-Seq using the VirScan library. Plots display normalized enrichment values for technical replicates for (**A**) positive control HIV-1 V3-directed gp120 antibody 447-52D, (**B-D**) QA013.2 lineage (inferred naïve precursor, intermediate, and mature), (**E**) negative isotype control (SARS-CoV-2 spike antibody C68.59 inferred naïve precursor). All antibodies were used at 10 ng/well. Data are presented using nominal p-values ($p < 0.05$ in both replicates) without correction for multiple hypothesis testing. This less stringent, exploratory approach was applied specifically to maximize sensitivity for detecting weak-binding targets. Each data point represents a peptide target meeting this nominal significant threshold for both replicates. The colors indicate peptides corresponding to the indicated virus. Benjamini–Hochberg corrected hit lists are reported in TableS7.

linear amino acid motif was apparent among them, suggesting that these shared hits do not represent a coherent motif-driven specificity. We further searched all hits with $p < 0.05$ for peptides containing the GDIR motif, given that mature QA013.2 partially depends on this motif for HIV Env binding. No such peptides were identified, despite confirmation that GDIR-containing peptides were present in the library.

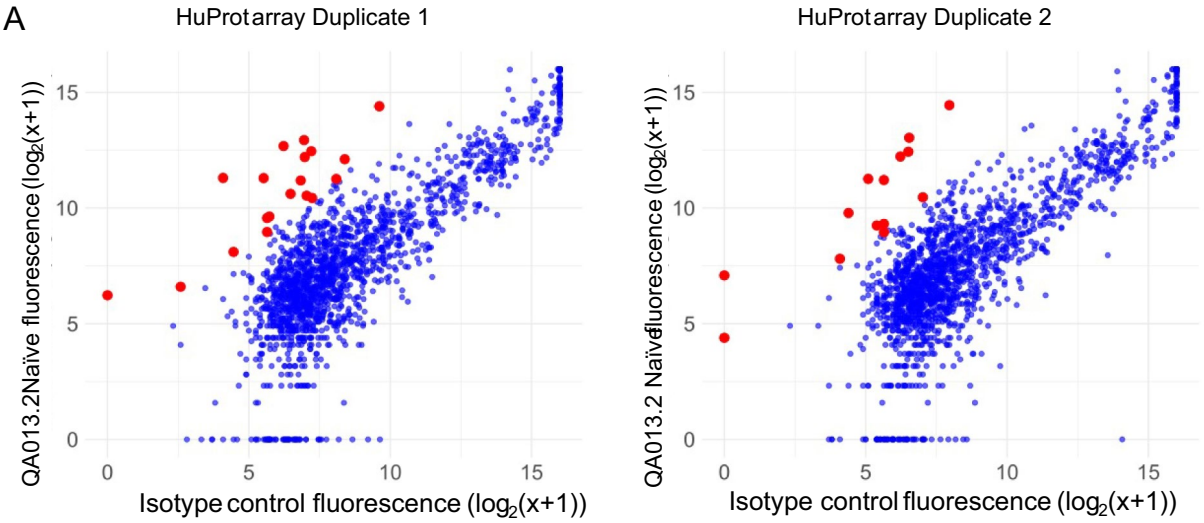
Together, these data suggest that QA013.2 antibodies do not exhibit strong or consistent cross-reactivity to non-HIV-derived viral linear peptides in the VirScan library.

## Human protein microarray did not reveal a compelling dominant target for the inferred naïve precursor QA013.2 antibody

To assess whether inferred naïve precursor of QA013.2 recognized candidate human proteins, we screened the antibody on the HuProt human protein microarray, a proteome-scale platform containing more than 21,000 human proteins and isoform variants. Binding was evaluated relative to an isotype control, reactivity was identified for each duplicate spot using a linear regression model of QA013.2 inferred naïve precursor versus isotype control binding, followed by residual analysis (Fig. 6A). Proteins were classified as candidate hits if they were positive residual outliers according to the $1.5 \times$ IQR rule, indicating higher than expected QA013.2 binding relative to the isotype control.

This analysis identified several candidate protein reactivities (Fig. 6B, TableS8), with the strongest signal observed for RFK (21,934 for QA013.2 vs 516 for isotype control). Additional hits included SMARCE1 (8121 vs 108), ABHD4 (5574 vs 119), HMCES (4754 vs 100), CRIP2 (2486 vs 25), and NSDHL (2441 vs 47). COX4I2 was also detected, but at much lower intensity (105 vs 0). Despite good concordance between duplicates and differential binding intensities, the candidate proteins were heterogeneous and lacked an obvious shared motif. Furthermore, their localizations span distinct intracellular compartments—ranging from nuclear (HMCES, SMARCE1) and cytosolic proteins (RFK, CRIP2) to internal organellar membrane proteins (ABHD4, NSDHL, COX4I2)—which is difficult to reconcile with a plausible priming antigen. None possess domains or signal sequences required for secretion or surface expression[31], and while processes such as cell death or extracellular vesicle release can expose intracellular epitopes externally, the role of these potential routes and antigens in priming the QA013.2 lineage remains unconfirmed. Thus, although the array identified a small set of reproducibly enriched protein reactivities, it did not point to a convincing dominant target, and we interpreted these HuProt hits conservatively as provisional protein reactivities rather than evidence for a defined precursor antigen.

A




B

| Protein Symbol | Uniprot | Protein name | Mean $log_2$ FC | QA013.2 Naïve Raw dup1 | Isotype Raw dup1 | QA013.2 Naïve Raw dup2 | Isotype Raw dup2 |
|---|---|---|---|---|---|---|---|
| CRIP2 | P52943 | Cysteine-rich protein 2 | 6.7 | 2523 | 16 | 2448 | 33 |
| COX4I2 | Q96KJ9 | Cytochrome c oxidase subunit 4 isoform 2 | 6.6 | 74 | 0 | 135 | 0 |
| SMARCE1 | Q969G3 | SWI/SNF-related, matrix-associated, actin-dependent regulator of chromatin subfamily E member 1 | 6.2 | 7832 | 123 | 8409 | 92 |
| NSDHL | Q15738 | NAD(P)-dependent steroid dehydrogenase-like | 5.7 | 2511 | 45 | 2370 | 49 |
| ABHD4 | Q8TB40 | Abhydrolase domain-containing protein 4 | 5.6 | 5620 | 147 | 5528 | 90 |
| HMCES | Q96FZ2 | 5-hydroxymethylcytosine binding, ES-cell-specific protein | 5.6 | 4717 | 125 | 4790 | 74 |
| RFK | Q969G6 | Riboflavin kinase | 5.6 | 21590 | 783 | 22278 | 248 |

**Fig. 6** Assessment of QA013.2 inferred naïve precursor antibody binding to human proteome. (**A**) Scatter plot showing QA013.2 inferred naïve precursor (1 µg/mL) binding HuProt microarray proteins. Each point represents an individual protein. To account for zero values, fluorescence intensities were $log_2$ (x + 1) transformed. The x-axis and y-axis show $log_2$(x + 1) transformed fluorescence intensity for the isotype control antibody (SARS-CoV-2 spike antibody C68.59 inferred naïve precursor, 1 ug/mL), and for QA013.2 inferred naïve precursor antibody, respectively (1 µg/mL). To compare the pairwise relation between two arrays (QA013.2 vs isotype), a linear regression model was fitted and residuals were calculated. Outliers were defined using the 1.5 × interquartile range (IQR) method applied to the residuals, and only proteins with positive residuals (enriched binding to QA013.2 inferred naïve precursor relative to isotype) were retained. These proteins are highlighted in red. (**B**) Table summarizing proteins with consistently high binding to QA013.2 inferred naïve precursor across both duplicates, defined as those identified as positive-residual outliers in both datasets. The table reports the mean $log_2$ fold change as well as raw fluorescence intensities for each duplicate (dup1 and dup2).

## Discussion

Our hypothesis that HIV bnAb precursors may recognize non-HIV antigens is supported by the observation that pre-HIV antibodies in the QA013.2 lineage had already acquired mutations later found in the mature bnAb before documented HIV infection. Previous studies also support this idea: some HIV-1 bnAbs can cross-react with non-HIV antigens, including Candida albicans, influenza hemagglutinin, lipooligosaccharide, and SARS-CoV-2 antigens[24,32–36]. Conversely, antibodies elicited outside the context of HIV infection can bind HIV antigens[37], and monoclonal antibodies isolated from HIV-naïve donors can recognize HIV gp41[21–23]. Together, these observations support the idea that pre-existing antigenic exposures may seed or shape lineages that later mature toward HIV recognition.

Here, we leveraged the QA013.2 bnAb lineage to test the reactivity of its inferred naïve precursor and an intermediate lineage antibody circulating pre-HIV infection against non-HIV antigens. We did not detect glycan reactivity for the inferred naïve precursor QA013.2 antibody on either the microbial glycan microarray or the schistosome-focused glycan array. Our negative glycan array results should be interpreted in the context of prior studies showing that some glycan-dependent HIV-1 bnAbs cross-react with non-HIV glycans, including parasite glycans from *Schistosoma mansoni*, and that the inferred naïve precursor of a N332/V3 bnAb lineage can bind *Schistosoma mansoni* cercariae despite lacking gp120 reactivity[24]. While these prior studies show that non-HIV glycans can engage early precursors of glycan-reactive bnAb lineages, our results suggest that QA013.2 may rely on distinct mechanisms of lineage initiation. Alternatively, an initiating glycan interaction may lie below the detection sensitivity of our platforms or depend on spatial contexts not recapitulated on the tested arrays. Furthermore, our ELISA evaluation of Schistosoma reactivity was limited to the egg antigen fraction, which may not capture stage-specific glycans expressed by cercariae or adult worms.

Likewise, VirScan profiling yielded no candidate viral peptides for the QA013.2 lineage. Although the platform successfully recovered expected HIV-1 peptide reactivity for a benchmark antibody, there were no peptides that bound antibodies in the QA013.2 lineage in an analysis that corrected for multiple comparisons. At a more permissive unadjusted nominal threshold, the candidate peptide enrichments that were identified were broad, lacked a conserved sequence motif, and were indistinguishable from non-specific reactivity observed in the isotype control. These results do not rule out that a heterologous viral antigen contributed to lineage initiation, but suggest that if so, it may not be represented as a linear epitope; rather it may require a conformational context or glycosylation. In general, there is little data to indicate what level of binding is needed for signaling of the B cell receptor and thus we also cannot rule out that binding below the sensitivity of this assay is nonetheless biologically relevant.

Similarly, the human proteome-scale screen did not identify a compelling dominant protein target for the inferred naïve precursor of the QA013.2 antibody. Although the HuProt array revealed several candidate protein hits with reproducible high binding, these hits were heterogeneous and did not point to a clear shared property, sequence or motif. Several candidates were also intracellular or otherwise difficult to reconcile with a plausible priming antigen. These signals could be a sign of an early precursor or simple cross-reactivity, as it is commonly found in monoclonal antibodies[38], suggesting that these proteins have some aspect that partially mimics the antigen sufficiently to bind with enough affinity to the microarray but not with high affinity. Thus, while these data do not exclude context-dependent protein reactivity, they do not provide strong evidence for a defined human protein precursor antigen for QA013.2.

## Conclusion

Collectively, we assessed the binding of the inferred naïve precursor of the QA013.2 antibody against more than 21,000 human proteins, 340 glycans, and 93,000 viral peptides. Although pre-HIV infection sequence data suggest that the QA013.2 lineage may have been initiated by non-HIV antigens, we were unable to identify the specific antigen(s). Importantly, these data should not be interpreted as evidence that no initiating heterologous antigen existed. This work makes an important contribution by evaluating a wide array of candidate antigens using a uniquely well-supported inferred naïve precursor antibody enabled by longitudinal sampling before HIV infection. More broadly, our findings highlight the current limitations of existing antigen discovery platforms. They suggest that the earliest triggers of bnAb lineage development may involve low-affinity interactions or may be transient, conformational, or dependent on modes of antigen presentation not captured by current high-throughput approaches. Defining these earliest antigenic triggers remains a major unresolved problem in HIV immunology and an important challenge for future vaccine design.

## Limitations

Our approach is constrained by platform-specific limitations in epitope display and sensitivity. VirScan is limited to the detection of linear non-glycosylated peptides; therefore, conformational epitopes and native viral modifications may have been missed. Similarly, HuProt proteins produced in yeast may lack mammalian folding or post-translational modifications and glycosylations. Furthermore, static, isolated antigen displays on these arrays lack the high valency, membrane or follicular dendritic cell display features, and cannot capture transient antigen exposure. For these assays, as with other binding assays, the relationship between the detection threshold used to define binding and the level of binding required for cellular signaling remains poorly defined. Thus, inferred naïve precursor antibody binding may fall below the detection limit of our assays, although inclusion of intermediate and mature antibody variants was intended to mitigate this.

Because QA013.2 sequence identification relied on bulk sequencing[19], heavy- and light-chain pairing remains computationally inferred rather than validated at single-cell resolution, presenting a potential source of uncertainty for antigen interactions. Furthermore, computational inference inherently carries limitations. However, exceptionally deep sampling across pre-HIV infection timepoints and state-of-the-art methods[39] minimize sequence uncertainty across the inferred lineage. Accordingly, empirical binding characterization in this study focuses on a key intermediate along the primary evolutionary path, rather than exhaustive functional testing of all potential intermediate variants. Because our primary objective was to identify heterologous antigens capable of driving early lineage initiation, we prioritized inferred naïve precursor and pre-HIV intermediate across all screening platforms. Thus, our study was not designed to address the evolution of the response to the antigens sampled in our assay.

## Methods

### Monoclonal antibodies

The QA013.2 inferred naïve precursor, mature and intermediate antibody sequences as well as the SARS-CoV-2 antibody C68.59 inferred naïve precursor—which is used as an isotype control in this study—were retrieved from previously published studies[19,40]. In those prior studies, ancestral sequence reconstruction of the QA013.2 lineage was performed on bulk deep sequencing data generated from PBMC samples collected across multiple longitudinal timepoints, including pre- and post- HIV infection samples[19]. To infer the naïve precursor and intermediates along the developmental pathway, clonal clustering and V(D)J germline gene assignments were determined using *partis*, followed by Bayesian phylogenetic modeling in *linearham*[41]. This framework is a phylogenetic hidden Markov model that accounts for both VDJ recombination and SHM when inferring ancestral sequences including the inferred naïve B cell receptor (BCR), resulting in a posterior distribution over naïve sequences, which can be viewed for example in Fig. 1 of Shipley et al.[19]. From these previously published data, the QA013.2 intermediate (VH6 and VL5 sequences) was selected for the current study as it represents a key intermediate stage in the QA013.2 lineage maturation, exhibiting 6% total SHM relative to the inferred naïve precursor, and because it corresponds to sequences detected with 99–100% identity in bulk sequencing 379 days prior to HIV infection, thus reducing the sequence uncertainty inherent to computational inferences. Despite this maturation, QA013.2 intermediate does not neutralize autologous or heterologous HIV strains and does not bind to the autologous superinfecting virus and showed weak affinity for the heterologous BG505 SOSIP[19]. QA013.2 and isotype control antibody (C68.59 inferred naïve precursor) sequences used in this study are available in TableS1.

Antibodies were synthesized with standard methods previously described[40]. Briefly, gene fragments for heavy and light chains were ordered from Twist Bioscience and cloned into Igγ1, Igλ expression vectors[42]. Equal ratios of heavy and light chain plasmids were co-transfected into 293 F cells using FreeStyle MAX (Thermo Fisher Scientific, catalog #16,447,100) according to the manufacturer's instructions. FreeStyle 293 F cells (Thermo

Fisher Scientific #R790-07) were grown at 37 °C in Freestyle 293 Expression Media (Thermo Fisher Scientific) according to the manufacturer's instructions. Column-based Pierce protein G (Thermo Fisher Scientific, catalog #20,397) purification of IgG was done according to the manufacturer instructions.

The following HIV-1 gp120 monoclonal antibodies were obtained through the National Institute of Allergy and Infectious Diseases (NIAID) HIV Reagent Program (Division of AIDS, National Institutes of Health): VRC01 (ARP-12033; via BEI Resources), PGT121 (ARP-12343; contributed by the International AIDS Vaccine Initiative), and 447-52D (ARP-4030; contributed by Dr. Susan Zolla-Pazner).

All monoclonal antibodies used in this study were of the IgG1 isotype and evaluated at equivalent concentrations within each assay.

## Glycan microarrays

Glycan binding was evaluated using a Microbial Glycan Microarray [29] and the Schistosome glycan microarray[43,44] provided by the Consortium for Functional Glycomics and National Center for Functional Glycomics (NCFG). The printed array consists of polysaccharides derived from 313 different Gram-negative and Gram-positive bacterial species, each printed at 500 µg/mL in six technical replicates. The Schistosome array consists of 32 compounds (glycans and glycopeptides, and undefined fractions from *Schistosoma* eggs). Further detailed information on the arrays composition and glycan structures is available through the NCFG [29,44].

Slides were removed from 4 °C storage, rinsed with distilled water, and briefly centrifuged to dry before application of a SecureSeal hybridization chamber. Slides were rehydrated for 5 min with wash buffer (TSM Buffer + 0.05% Tween-20; TSM composition: 20 mM Tris–HCl pH 7.4, 150 mM NaCl, 2 mM $CaCl_2$, and 2 mM $MgCl_2$). Antibodies were diluted to 50 µg/mL in 400 µL binding buffer (TSM buffer + 0.05% Tween-20 + 1% BSA) and incubated on the slide for 1 h at room temperature with gentle agitation. Slides were then washed four times with wash buffer followed by four washes with TSM buffer. For secondary detection, goat anti-human IgG biotin-conjugated antibody (Jackson ImmunoResearch, Cat. No. 109–065-088) was diluted to 5 µg/mL in binding buffer (1:400 final dilution in 400 µL total volume; TSM buffer + 0.05% Tween-20 + 1% BSA) and incubated for 1 h at room temperature with shaking. Slides were washed four times with wash buffer and four times with TSM buffer. For signal detection, Cy5-streptavidin (Thermo Fisher, Cat. No. 434316) was diluted in binding buffer to a final volume of 1000 µL (0.5 µg Cy5-streptavidin in TSM buffer). Slides were incubated for 1 h at room temperature in the dark with agitation, followed by four washes with wash buffer and four washes with TSM buffer. Slides were scanned at 635 nm using the ImageXpress Pico system (Molecular Devices) with an exposure time of 500 ms. Fluorescence intensities were extracted using GenePix Pro 7.4 software (Molecular Devices) and were reported as relative fluorescence units (RFU). Control serum from anonymous healthy adult donors (18–65 years old, Boston area) was included to illustrate the magnitude of positive binding signals on the microbial glycan microarray. For the Schistosoma glycan array, a positive-control lectin cocktail comprising Concanavalin A (ConA; 10 µg/mL, Vector Labs), Aleuria aurantia lectin (AAL; 1 µg/mL, Vector Labs), and Wisteria floribunda agglutinin (WFA; 10 µg/mL, Vector Labs) was included; these lectins recognize high-mannose/N-linked glycans, fucosylated glycans, and terminal N-acetylgalactosamine (GalNAc)-containing glycans, respectively. Data were analyzed using the MAWE method as described previously[45]. For the microbial glycan array, the highest and lowest values from each set of six replicates were removed to eliminate the false hits that contain a single very high or low point. RFU values represent background-subtracted fluorescence intensities averaged across the remaining replicate spots. No fixed absolute RFU threshold was applied; rather, detectable binding was defined as a reproducible elevation above the local background and negative-control signal. Background was assessed using the C68.59 inferred naïve precursor antibody as a negative control, which showed RFU ranges of -1.00 to 2.25 on the microbial glycan microarray and -0.25 to 1.75 on the Schistosoma glycan array. Glycans were considered reactive only if signals exceeded a reproducible elevation above the magnitude and distribution of the negative isotype control signal across replicate spots; signals comparable to the negative-control distribution or lacking reproducibility were classified as no detectable binding.

## ELISA

Human anti-*Schistosoma mansoni* IgG ELISA kit (abcam Cat ab108769) was used to assess reactivity of plasma and antibodies as per manufacturer recommendations. Plasma samples diluted to 1:100 and antibodies used at 50 µg/mL. Each sample was run in technical duplicate for every independent repeat. Absorbance (OD450) was read using an Epoch reader (software v2.06.10). Standard units were calculated for each experiment by normalizing the blank-corrected sample OD to the blank-corrected cutoff-control OD and multiplying by 10 according to the manufacturer instructions. Values from the two independent experiments were pooled for plotting.

## VirScan

VirScan was performed using the T7 VirScan v3.0 bacteriophage library, kindly provided by Stephen Elledge, which displays tiled peptides spanning viruses with human tropism as previously described[46,47]. Samples were normalized to 10 ng IgG per well. The library was diluted to approximately 2.5 × 10^10 pfu in a 1 mL reaction and incubated with sample IgG for 20 h at 4 °C with gentle rocking. Immune complexes were captured with 40 µL of a 1:1 mixture of Protein A and Protein G Dynabeads for 4 h at 4 °C, followed by three washes in wash buffer. Bound phage DNA was recovered by heating the bead suspension at 95 °C for 10 min. Recovered inserts were PCR-amplified, indexed, pooled, gel-purified, and sequenced on an Illumina NextSeq P1 using 75-cycle single-end reads. Mock immunoprecipitations and library-only input controls were included on each plate, and all samples were analyzed in duplicate.

Sequencing counts were normalized and analyzed using the phippery pipeline[48] (phip-flow V1.14 and phippery 1.3.1). Background variability is modeled by fitting a negative binomial to the beads-only samples to obtain a per-peptide dispersion, which is then held fixed when each sample is compared against the beads-only controls. Two metrics were computed for each peptide in each sample: (i) a p value assessing whether counts in the empirical sample were significantly above background, reported as -log10(p), and (ii) enrichment was calculated as the fold change in frequency (counts per million) of a given peptide within each sample compared to the input library-only control; a pseudocount of 1 is applied in the enrichment calculation. Two complementary significance criteria were used. In the stringent analysis, peptides were classified as hits if they reached Benjamini-Hochberg-adjusted $p < 0.05$ in both technical replicates. In a separate exploratory analysis designed to identify potential weaker binders, peptides were considered "provisional" hits if they reached a nominal $p < 0.05$ in both technical replicates, without correction for multiple testing.

## HuProt

HuProt Human Proteome Microarray v4.0 slides (CDI Laboratories) were stored at −80 °C and kept covered on dry ice until use to minimize condensation. Arrays were blocked in 5% BSA in 1 × TBS-T for 1.5 h at room temperature with shaking at 40 rpm. For primary incubation, blocking buffer was removed and arrays were incubated with 3 µg monoclonal antibody in 3 mL blocking buffer for 1 h at room temperature. Slides were then washed three times in 1 × TBS-T for 10 min each with gentle shaking. For detection, arrays were incubated with anti-human IgG Alexa Fluor 647 (Jackson ImmunoResearch, 709–605-149) diluted 1:1000 in 5% BSA in 1 × TBS-T for 1 h at room temperature in the dark. After three additional 10-min washes in 1 × TBS-T, arrays were rinsed in water and prepared for scanning.

Microarray images were analyzed using GenePix Pro 6.1 (Molecular Devices) to obtain spot-level fluorescence intensities. For each feature, foreground median intensity and local background median intensity were extracted, and background-corrected fluorescence was calculated as F635 median minus B635 median. Raw fluorescence intensities were log2-transformed (log2[x + 1]) to stabilize variance and reduce the impact of extreme values. Spots with non-missing values for both antibodies were retained, and duplicate spots were analyzed separately to allow detection of reproducible reactivity across duplicates. For each duplicate (dup1, dup2) a linear

model was fitted to describe the relationship between isotype control signal (SARS-CoV-2 spike antibody C68.59 inferred naïve precursor) and antibody-specific signal (QA013.2 inferred naïve precursor). Residuals from this model were used as a measure of deviation from expected background binding patterns. Enriched proteins were identified as outliers based on the interquartile range (IQR) criterion applied to model residuals (1.5 × IQR beyond the first or third quartile), and only proteins showing positive residuals (enrichment in QA013.2 inferred naïve precursor relative to isotype control) were retained. Fold change was calculated in log2 scale (log2(QA013.2 inferred naïve precursor) – log2(isotype control)). Proteins identified as positive residual outliers in both duplicate sets were considered reproducible candidate hits. Data visualization was performed in R 4.5.2.

## Additional quantification and statistical analysis

Additional quantification and statistical analyses were performed in Prism (GraphPad). Plots were made using R and Prism. Additional sequence data analysis was conducted using Geneious. Figure 2 was generated with BioRender.

## Ethics and human subjects

This study was approved by members of the ethical review committees at the University of Nairobi, the Fred Hutchinson Cancer Research Center, and the University of Washington. Study participants provided written informed consent prior to enrollment. Plasma samples were collected from female participant QA013 at various times pre- and post-HIV infection. QA013 was enrolled in a prospective cohort of HIV-negative, high-risk women in Mombasa, Kenya[49]. Study participants were treated according to Kenyan National Guidelines and did not receive any antiretroviral treatment during the time period in which the samples used in this study were collected. Antiretroviral therapy was offered as of March 2004, with support from the President's Emergency Plan for AIDS Relief.

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**Author contributions** JO, MS and DMD conceptualized the project. DMD performed data collection, analysis, interpretation and visualization. FAM and KS analyzed VirScan data; LAB acquired HuProt data. YZ and PL assisted with HuProt analysis and interpretation. The manuscript was written by DMD and JO and critically revised by all authors.

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**Data availability** VirScan raw sequencing data are available at NCBI SRA (PRJNA1463238), VirScan processed data at Zenodo (https://doi.org/10.5281/zenodo.20033894) and analysis code on GitHub (https://github.com/matsengrp/QA013.2-HIV-bnAb-analysis). All raw data and figure source data are provided in supplementary tables.

## Declarations

**Competing interests** The authors declare no competing interests.

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